1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · High difficulty / trap

P06 — Inverse problem: rebuild the matrix of a spiral sink from one real solution, solve an IVP and measure the decay per turn

The real 2×2 matrix A of the linear system $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x} = (x, y)^T$, is unknown. The only information available is that the system has the real solution

$$\mathbf{x}_1(t) = e^{-t} \begin{pmatrix} \cos 2t - \sin 2t \\ \cos 2t - 2\sin 2t \end{pmatrix}.$$

(a) Reconstruct A. Show that its eigenvalues are $-1 \pm 2i$, find a complex eigenvector, and check that $P^{-1}AP$ is the real canonical form, with P the matrix whose columns are the real and imaginary parts of that eigenvector, normalised so that the given solution is $\mathrm{Re}(e^{\lambda t}\boldsymbol{\eta})$.

(b) Find a second real solution, prove that the two solutions form a fundamental set, write the real general solution and solve the initial value problem $\mathbf{x}(0) = (1, 0)^T$.

(c) Classify the origin. Do the trajectories spiral in or out? Is the rotation clockwise or counter-clockwise? Justify with a test vector.

(d) (extension: the quadratic form V is not in the course; it follows from the canonical form.) Let $V(x, y) = 5x^2 - 6xy + 2y^2$. Show that $V(\mathbf{x}) = |P^{-1}\mathbf{x}|^2$ and that $\frac{d}{dt}V(\mathbf{x}(t)) = -2V(\mathbf{x}(t))$ along every solution. For the solution of (b), find the time needed for $\sqrt{V}$ to halve, and the first time $t > 0$ at which the solution is back on the positive x axis, with the ratio of its distances to the origin there and at $t = 0$. Is the Euclidean distance $|\mathbf{x}(t)|$ decreasing for every solution?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: the four real entries of A; then the solutions $\mathbf{x}(t) \in \mathbb{R}^2$, $t \in \mathbb{R}$.
- Operator: first-order, linear, homogeneous, constant coefficients, planar. Solutions exist for all t and are unique for given $\mathbf{x}(0)$ (existence and uniqueness for linear systems).
- Datum: one solution $\mathbf{x}_1$. Since $\mathbf{x}_1'(t) = A\mathbf{x}_1(t)$ for every t, evaluating at two instants where $\mathbf{x}_1$ takes independent values gives A on a basis, hence A.
- Course convention (Robinson Ch. 29): if $\lambda = \rho + i\omega$ ($\omega > 0$) has eigenvector $\boldsymbol{\eta} = \mathbf{u} + i\mathbf{w}$, the real solutions are $e^{\rho t}(\cos \omega t\, \mathbf{u} - \sin \omega t\, \mathbf{w})$ and $e^{\rho t}(\sin \omega t\, \mathbf{u} + \cos \omega t\, \mathbf{w})$.

Quantity	Value
$\mathbf{x}_1(0)$	$(1, 1)^T$

Quantity	Value
$\mathbf{x}_1(\pi/4)$	$e^{-\pi/4}(-1,-2)^T$
Initial condition (b)	$\mathbf{x}(0)=(1,0)^T$

Phase 2. Canonical formulation

Matrix form $\mathbf{x}'=A\mathbf{x}$. The target canonical form ("frame" of the problem) is $\mathbf{z}'=J\mathbf{z}$ with $\mathbf{x}=P\mathbf{z}$, $P=[\mathbf{u}\mid\mathbf{w}]$ and $J=\begin{pmatrix}\rho & \omega\\ -\omega & \rho\end{pmatrix}$, so that $A=PJP^{-1}$.

Part (a) · 1.5 points

Reconstruct A, its eigenvalues and eigenvector, and check the canonical form

Step 1. Derivative of the given solution

Write $c=\cos 2t$, $s=\sin 2t$. By the chain rule $\frac{d}{dt}\cos 2t=-2\sin 2t$, $\frac{d}{dt}\sin 2t=2\cos 2t$, and by the product rule $\frac{d}{dt}\big(e^{-t}g(t)\big)=-e^{-t}g(t)+e^{-t}g'(t)$:

$$\mathbf{x}_1'(t)=-e^{-t}\begin{pmatrix}c-s\\ c-2s\end{pmatrix}+e^{-t}\begin{pmatrix}-2s-2c\\ -2s-4c\end{pmatrix}=e^{-t}\begin{pmatrix}-3c-s\\ -5c\end{pmatrix}$$

Step 2. Two instants give A on a basis

At $t=0$ ($c=1,s=0$): $\mathbf{x}_1(0)=(1,1)^T$, $\mathbf{x}_1'(0)=(-3,-5)^T$. At $t=\pi/4$ ($c=0,s=1$): $\mathbf{x}_1(\pi/4)=e^{-\pi/4}(-1,-2)^T$, $\mathbf{x}_1'(\pi/4)=e^{-\pi/4}(-1,0)^T$. Dividing the second relation by $-e^{-\pi/4}$:

$$A\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}-3\\-5\end{pmatrix},\qquad A\begin{pmatrix}1\\2\end{pmatrix}=\begin{pmatrix}1\\0\end{pmatrix}$$

With $P=\begin{pmatrix}1&1\\1&2\end{pmatrix}$ this reads $AP=\begin{pmatrix}-3&1\\-5&0\end{pmatrix}$. $\det P=1\cdot 2-1\cdot 1=1\neq 0$, and $P^{-1}=\frac{1}{1}\begin{pmatrix}2&-1\\-1&1\end{pmatrix}$, so $A=(AP)P^{-1}$ entry by entry:

$$\begin{aligned}A_{11}&=(-3)(2)+(1)(-1)=-7, & A_{12}&=(-3)(-1)+(1)(1)=4,\\ A_{21}&=(-5)(2)+(0)(-1)=-10, & A_{22}&=(-5)(-1)+(0)(1)=5.\end{aligned}$$

$$A = \begin{pmatrix} -7 & 4 \\ -10 & 5 \end{pmatrix}$$

Step 3. Eigenvalues

$$p(\lambda) = \det(A - \lambda I) = (-7-\lambda)(5-\lambda) - (4)(-10) = \lambda^2 + 2\lambda - 35 + 40 = \lambda^2 + 2\lambda + 5$$

$\lambda = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$, so $\rho = -1, \omega = 2$. Check: $\operatorname{tr} A = -2 = 2\rho$, $\det A = -35 + 40 = 5 = \rho^2 + \omega^2$.

Step 4. Eigenvector from $(A - \lambda I)\boldsymbol{\eta} = \mathbf{0}$

For $\lambda = -1 + 2i$, row 1: $(-7 - (-1+2i))\eta_1 + 4\eta_2 = 0$, i.e. $(-6 - 2i)\eta_1 + 4\eta_2 = 0$, so $\eta_2 = \frac{3+i}{2}\eta_1$. Taking $\eta_1 = 1 + i$: $\eta_2 = \frac{(3+i)(1+i)}{2} = \frac{3+3i+i-1}{2} = 1 + 2i$. Row 2 as a check: $-10(1+i) + (5 - (-1+2i))(1+2i) = -10 - 10i + (6-2i)(1+2i) = -10 - 10i + 6 + 12i - 2i + 4 = 0$. Hence

$$\boldsymbol{\eta} = \begin{pmatrix} 1+i \\ 1+2i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\mathbf{u}} + i \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{\mathbf{w}}$$

and indeed $\mathbf{x}_1 = e^{-t}(\cos 2t\,\mathbf{u} - \sin 2t\,\mathbf{w})$: the given curve is the real part of $e^{\lambda t}\boldsymbol{\eta}$.

Step 5. Frame check $P^{-1}AP = J$

$P = [\mathbf{u} \mid \mathbf{w}]$ is the matrix of Step 2. With $AP = \begin{pmatrix} -3 & 1 \\ -5 & 0 \end{pmatrix}$:

$$\begin{aligned} (P^{-1}AP)_{11} &= 2(-3) + (-1)(-5) = -1, \quad (P^{-1}AP)_{12} = 2(1) + (-1)(0) = 2, \\ (P^{-1}AP)_{21} &= (-1)(-3) + 1(-5) = -2, \quad (P^{-1}AP)_{22} = (-1)(1) + 1(0) = -1. \end{aligned}$$

$$P^{-1}AP = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix} = J$$

KEY RESULTS

- $A = \begin{pmatrix} -7 & 4 \\ -10 & 5 \end{pmatrix}, \operatorname{tr} A = -2, \det A = 5, p(\lambda) = \lambda^2 + 2\lambda + 5, \lambda = -1 \pm 2i.$
- $\boldsymbol{\eta} = (1+i,\ 1+2i)^T, \mathbf{u} = (1,1)^T, \mathbf{w} = (1,2)^T.$
- $P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \det P = 1, P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, P^{-1}AP = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}.$

Part (b) · 1.5 points

Second real solution, fundamental set, general solution and IVP

Step 1. Complex solution split into real and imaginary parts

With Euler's formula $e^{(-1+2i)t} = e^{-t}(\cos 2t + i \sin 2t)$:

$$e^{\lambda t}\boldsymbol{\eta} = e^{-t}(\cos 2t + i \sin 2t)(\mathbf{u} + i\mathbf{w}) = e^{-t}\big[(\cos 2t\,\mathbf{u} - \sin 2t\,\mathbf{w}) + i(\sin 2t\,\mathbf{u} + \cos 2t\,\mathbf{w})\big]$$

Since A is real, real and imaginary parts are real solutions (Robinson Ch. 29). The real part is $\mathbf{x}_1$; the imaginary part is

$$\mathbf{x}_2(t) = e^{-t}(\sin 2t\,\mathbf{u} + \cos 2t\,\mathbf{w}) = e^{-t}\begin{pmatrix} \sin 2t + \cos 2t \\ \sin 2t + 2\cos 2t \end{pmatrix}$$

Step 2. Fundamental set

$\mathbf{x}_1(0) = \mathbf{u}, \mathbf{x}_2(0) = \mathbf{w}$, so the Wronskian at $t = 0$ is $W(0) = \det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = 2 - 1 = 1 \neq 0$. $W(0) \neq 0$ is enough for independence (Notes, p. 9). Abel's formula for systems (**extension**) adds $W(t) = W(0)e^{(\operatorname{tr} A)t} = e^{-2t} \neq 0$ for all t. Hence the general real solution is

$$\mathbf{x}(t) = c_1 e^{-t}\begin{pmatrix} \cos 2t - \sin 2t \\ \cos 2t - 2\sin 2t \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} \sin 2t + \cos 2t \\ \sin 2t + 2\cos 2t \end{pmatrix}, \qquad c_1, c_2 \in \mathbb{R}$$

Step 3. Initial condition

At $t=0$: $c_1\mathbf{u}+c_2\mathbf{w}=(1,0)^T$, i.e. $c_1+c_2=1$ and $c_1+2c_2=0$. Subtracting, $c_2=-1$, then $c_1=2$:

$$\mathbf{x}(t)=e^{-t}\begin{pmatrix}2(\cos 2t-\sin 2t)-(\sin 2t+\cos 2t)\\2(\cos 2t-2\sin 2t)-(\sin 2t+2\cos 2t)\end{pmatrix}=e^{-t}\begin{pmatrix}\cos 2t-3\sin 2t\\-5\sin 2t\end{pmatrix}$$

Step 4. Check by substitution

$\mathbf{x}(0)=(1,0)^T$. With $c=\cos 2t$, $s=\sin 2t$, the product rule gives $\mathbf{x}'=e^{-t}\big(-(c-3s)+(-2s-6c),\ 5s-10c\big)^T=e^{-t}(-7c+s,\ -10c+5s)^T$, and $A\mathbf{x}=e^{-t}\big(-7(c-3s)+4(-5s),\ -10(c-3s)+5(-5s)\big)^T=e^{-t}(-7c+s,\ -10c+5s)^T$. They agree.

KEY RESULTS

- $\mathbf{x}_2(t)=e^{-t}\begin{pmatrix}\sin 2t+\cos 2t\\\sin 2t+2\cos 2t\end{pmatrix}$, $W(0)=1\neq 0$.
- $\mathbf{x}(t)=c_1\mathbf{x}_1(t)+c_2\mathbf{x}_2(t)$; IVP: $c_1=2$, $c_2=-1$.
- $\mathbf{x}(t)=e^{-t}\begin{pmatrix}\cos 2t-3\sin 2t\\-5\sin 2t\end{pmatrix}$.

Part (c) · 1 point

Classification, spiralling and sense of rotation

Step 1. Type

The eigenvalues are complex with negative real part, so every solution is e^{-t} times a bounded π-periodic vector: it tends to $\mathbf{0}$ while winding around it. The origin is a stable focus (Robinson Ch. 31).

Step 2. Test vector

At $(1,0)$, on the positive x axis, $\mathbf{x}'=A(1,0)^T=(-7,-10)^T$: the y component is negative, so the solution crosses the positive x axis downwards. That is a **clockwise** rotation.

Step 3. Global confirmation

With θ the polar angle, $\theta'=\dfrac{xy'-yx'}{x^2+y^2}$. Here

$$xy'-yx'=x(-10x+5y)-y(-7x+4y)=-10x^2+12xy-4y^2=-2(5x^2-6xy+2y^2).$$

The quadratic form $5x^2-6xy+2y^2$ has matrix $\begin{pmatrix}5&-3\\-3&2\end{pmatrix}$ with $5>0$ and determinant $10-9=1>0$: it is positive definite. So $\theta'<0$ at every point except the origin: the rotation is clockwise everywhere. In the canonical plane J with $\omega=2>0$ rotates clockwise ($z_2'=-2z_1<0$ at $(1,0)$), and $\det P=1>0$

preserves orientation, which is consistent.

Step 4. Portrait in words

Trajectories are spirals that wind clockwise into the origin, one full turn every $T = 2\pi/\omega = \pi$ time units. They are not circles: they are images under P of logarithmic spirals of the $\mathbf{z}$ plane, so $\mathbf{u} = (1, 1)$ and $\mathbf{w} = (1, 2)$ are conjugate semi-diameters of the ellipses $V = \text{const}$ that the spirals cross.

KEY RESULTS

- $\rho = -1 < 0, \omega = 2 \neq 0$: **stable focus** (spiral sink); $\operatorname{tr} A = -2 < 0$, $\det A = 5 > 0$, $(\operatorname{tr} A)^2 - 4 \det A = -16 < 0$.
- $A(1, 0)^T = (-7, -10)^T$: **clockwise** rotation.
- $xy' - yx' = -2(5x^2 - 6xy + 2y^2) < 0$ off the origin.

Part (d) · 1 point

Decay of the canonical radius, return to the axis, Euclidean distance

Step 1. V is the canonical radius squared

$P^{-1}\mathbf{x} = \begin{pmatrix} 2x - y \\ -x + y \end{pmatrix}$, so $|P^{-1}\mathbf{x}|^2 = (4x^2 - 4xy + y^2) + (x^2 - 2xy + y^2) = 5x^2 - 6xy + 2y^2 = V$.

Step 2. Derivative along solutions

$V_x = 10x - 6y$, $V_y = -6x + 4y$, $x' = -7x + 4y$, $y' = -10x + 5y$. By the chain rule

$$\begin{aligned} \frac{dV}{dt} &= (10x - 6y)(-7x + 4y) + (-6x + 4y)(-10x + 5y) \\ &= (-70x^2 + 82xy - 24y^2) + (60x^2 - 70xy + 20y^2) = -10x^2 + 12xy - 4y^2 = -2V. \end{aligned}$$

(Equivalently $\frac{d}{dt}|\mathbf{z}|^2 = 2\mathbf{z}^T J \mathbf{z} = 2\rho|\mathbf{z}|^2$, because the skew part of J gives $\mathbf{z}^T \begin{pmatrix} 0 & \omega \\ -\omega & 0 \end{pmatrix} \mathbf{z} = 0$.)

Step 3. Integration

For $V \neq 0$ separate variables: $\displaystyle\int_{V(0)}^{V(t)} \frac{dV}{V} = \int_0^t (-2)\, d\tau$, so by Barrow's rule $[\ln V]_{V(0)}^{V(t)} = [-2\tau]_0^t$, i.e. $\ln V(t) - \ln V(0) = -2t$ and $V(t) = V(0)e^{-2t}$. For the IVP $V(0) = 5 \cdot 1^2 - 0 + 0 = 5$, so $V(t) = 5e^{-2t}$ and $\sqrt{V(t)} = \sqrt{5}\, e^{-t}$. Halving: $e^{-t_{1/2}} = \frac{1}{2}$, so $t_{1/2} = \ln 2 \approx 0.693$.

Step 4. Return to the positive x axis

$y(t) = -5e^{-t} \sin 2t = 0$ for $t = k\pi/2$. At $t = \pi/2$: $x = e^{-\pi/2} \cos \pi = -e^{-\pi/2} < 0$ (negative axis). At $t = \pi$: $x = e^{-\pi} \cos 2\pi = e^{-\pi} > 0$. So the first return is $t = \pi$, with $\mathbf{x}(\pi) = e^{-\pi}(1, 0)^T$ and ratio $|\mathbf{x}(\pi)|/|\mathbf{x}(0)| = e^{-\pi} \approx 0.0432$. This holds for every solution: the trigonometric factors have period $T = \pi$, so $\mathbf{x}(t + \pi) = e^{-\pi}\mathbf{x}(t)$.

Step 5. The Euclidean distance

$\frac{d}{dt}|\mathbf{x}|^2 = 2\mathbf{x}^T A\mathbf{x} = 2(-7x^2 + 4xy - 10xy + 5y^2) = 2(-7x^2 - 6xy + 5y^2)$. At $(0,1)$ it equals $10 > 0$: a solution through $(0,1)$ moves away from the origin at that instant. The Euclidean distance decays on average (by $e^{-\pi}$ per turn) but not monotonically; the quantity that decays monotonically, and exactly like e^{-t}, is $\sqrt{V} = |P^{-1}\mathbf{x}|$.

KEY RESULTS

- $V = |P^{-1}\mathbf{x}|^2 = 5x^2 - 6xy + 2y^2$, $\frac{dV}{dt} = -2V$, $V(t) = V(0)e^{-2t}$; for the IVP $V(t) = 5e^{-2t}$.
- $\sqrt{V}$ halves in $t_{1/2} = \ln 2 \approx 0.693$.
- Back on the positive x axis at $t = \pi$ (negative axis at $t = \pi/2$); $\mathbf{x}(\pi) = e^{-\pi}(1,0)^T$, ratio $e^{-\pi} \approx 0.0432$.
- No: $\frac{d}{dt}|\mathbf{x}|^2 = 2(-7x^2 - 6xy + 5y^2)$ is positive at $(0,1)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to +\infty$: $\mathbf{x} \to \mathbf{0}$ like e^{-t}; $t \to -\infty$: the spirals grow without bound. If ρ were 0 the trajectories would be the closed ellipses $V = \text{const}$ (a centre); here they cut those ellipses inwards.
- **Checks:** $A\mathbf{x}_1 = \mathbf{x}_1'$ for all t; $\operatorname{tr} A = 2\rho$, $\det A = \rho^2 + \omega^2$; $P^{-1}AP = J$; the IVP solution satisfies the system and $\mathbf{x}(0) = (1,0)^T$; Abel: $W(t) = e^{-2t}$.
- **Common mistakes in the exam:** (1) reading $\mathbf{w}$ with the wrong sign: the course writes the real part as $\cos\omega t\,\mathbf{u} - \sin\omega t\,\mathbf{w}$, so $\mathbf{w} = +(1,2)$; (2) deciding the sense of rotation from $\omega > 0$ alone: it must be checked with a test vector (it is clockwise here, and it would flip if $\det P < 0$); (3) claiming that the Euclidean distance halves after $\ln 2$: only the canonical radius $|P^{-1}\mathbf{x}|$ does; (4) taking the return time to the same half-axis as π/ω instead of $2\pi/\omega$.